Martin Chow

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Skills

Languages: Modern C++ (17/20), Python
Systems: Performance tuning, memory management, multithreading, cross-platform (Linux/Windows), diagnostics, profiling
Tools/Infra: CI/CD, Docker, Git, CMake, automated testing, developer tooling
Data/Visualization: 3D geometry processing, image pipelines, scientific visualization, full-stack tooling
Graphics/GPU: Vulkan, HLSL/GLSL, real-time ray tracing, compute shaders, GPU profiling (Nsight, RenderDoc)
AI/ML: PyTorch/TensorFlow (familiar), ML pipelines

Experience

Microsoft — SQL Performance Team
Software Developer *Jan 2022 – Jul 2023*

- Built performance automation tools and diagnostics workflows used across multiple engineering teams.
- Designed cross-platform testing infrastructure (Linux/Windows) to improve scalability and reliability.
- Developed CI/CD-integrated tooling for regression detection, profiling, and system-level performance analysis.
- Collaborated with senior engineers to improve developer experience and internal tooling efficiency.

University of Washington — DAIS Research Team
Software Engineer *Sep 2020 – Sep 2021*

- Developed Python-based 3D geometry and image-processing pipelines for scientific visualization.
- Built full-stack tools (Angular, Node.js, SQL) supporting research workflows, data inspection, and experiment management.
- Worked with researchers to translate domain-specific requirements into robust engineering solutions.
- Contributed to multidisciplinary projects involving visualization, data processing, and interactive tooling.

Projects

LuareEye Engine — Custom 3D Engine
Engine and Rendering Engineer *2025*

- Built a full 3D engine with animation systems, skeletal blending, and runtime asset pipelines.
- Designed a GPU-driven render pipeline with deferred lighting, render graphs, and multithreaded systems.
- Implemented real-time debugging tools, editor UI (IMGUI), and resource management systems.
- Used Nsight Graphics to analyze GPU workloads, debug pipeline transitions, and optimize performance.

Vulkan Real-Time Ray Tracing Engine — C++ Personal Project
Main Developer *2024*

- Implemented BLAS/TLAS acceleration structures, SBT pipelines, and ray tracing shader stages.
- Added reflections, soft shadows, GI, and SVGF denoising with TAA accumulation.
- Profiled GPU workloads using Nsight and RenderDoc to validate descriptor bindings and optimize shader performance.

Offline Ray Tracer — C++ Personal Project
Main Developer *2024*

- Built a CPU path tracer with importance sampling and multiple BRDF and IBL models based on SIGGRAPH research.

Lotus Child — Unity 2D Puzzle Game (24h Game Jam)
Core Gameplay Developer *2022*

- Built a complete 2D puzzle-platformer in 24 hours, including level design, character controller, and puzzle logic.
- Designed spatial logic challenges and implemented a lightweight interaction system.

Education

DigiPen Institute of Technology **Washington**
Master of Science in Computer Science *2024 – 2026 (expected)*